

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech (CE) Examination****April 2015****CE302/CE302.01 System Programming****Date: 25.04.2015, Saturday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections and both must be attempted in separate answer sheets.
2. Make suitable assumptions and draw neat figures wherever required.

SECTION – I**Q-1[A] Answer following. [06]**

(1) Define following terms.

(a) Semantic gap (b) System software (c) Pattern

(2) Explain language processing development tools.

[B] Answer the following. [06]

(1) What is language processor? List different flavors of language processor.

(2) Give Difference of Pass and Phase of a Language Processor.

(3) Differentiate Lexeme and Token.

Q-2[A] Answer the following. (Any two) [08]

(1) Which phases of compiler are part of front end and which are the part of back end?

(2) Differentiate single pass and multi pass compiler.

(3) Difference between scanner and parser.

[B] Draw DFA. (Any one) [04](1) $(a|b)^*ab$ (2) $(a|b)^*baaa$ **Q-3[A] Write regular expression. (Any three) [03]**

(1) String of 0 and 1 ending with 0

(2) String of 0 and 1 having next to last 0

(3) String of 0 and 1 having exactly two times 1

(4) String of 0 and 1 having first and last character different

[B] Answer the following. (Any two) [08]

(1) What is ambiguous grammar? Explain with example.

(2) Explain left recursion with example. Also explain how to remove it with example.

(3) What is parse tree? Draw a parse tree for given grammar and string.

Grammar : $S \rightarrow cAd$ String : cabd

$A \rightarrow ab|a$

(4) Write elements of assembly language with example.

SECTION – II

Q-4 Answer the following.

[12]

[A] Explain the data structure of single pass assembler.

[B] How to declare constant in assembly language?

[C] What is forward referencing? Explain with example.

Q-5[A] Answer the following. (Any two)

[08]

(1) What is editor? What are the functions of an editor?

(2) What are the methods available for debugging in Turbo C++?

(3) Explain the difference between relocatable and non-relocatable program.

(4) Why to use LTORG statement? What happens if LTORG statement is not used in program?

[B] Answer the following. (Any one)

[03]

(1) Differentiate compiler and interpreter.

(2) Mention at least 2 different ways to check the grammar is ambiguous

Q-6 Answer the following. (Any three)

[12]

[A] What is preprocessor and macro processor? Differentiate them.

[B] Explain single pass macro processor.

[C] Explain absolute loader.

[D] Write similarities and difference between macro and subroutine. How to decide whether to use a macro or a subroutine in program?

[E] Write a note on nested macro.
